

Advanced Program in **MERN STACK DEVELOPMENT**



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We assume you Might have some Questions

- Why should I choose krutanic?
- Why is Mern-Stack Development a good option?
- Do you know how many job opening's are worldwide?
- What are the Mern-Stack Development tool you need to master?
- Will I get a certificate to enhance my Resume?
- How should I Enroll for this program?
- How will the Mern-Stack Development Program will help you achieve your Dream?



We Genuinely care for your Dreams and be a path for your success!

- That's why our career focused programs seamlessly combine theory with practices, equipping you with real-world skills for employer's value
- Ensuring you're not just ready for the next step but also for the future at large, we offer exclusive modules on leveraging generative AI in your field



Get Placement Ready

Gain the most In demand skills to launch switch, or advance your career with actionable programs across fields



Community Building

Enjoy the best peer - led learning with tons of events while networking & building lifelong relationships



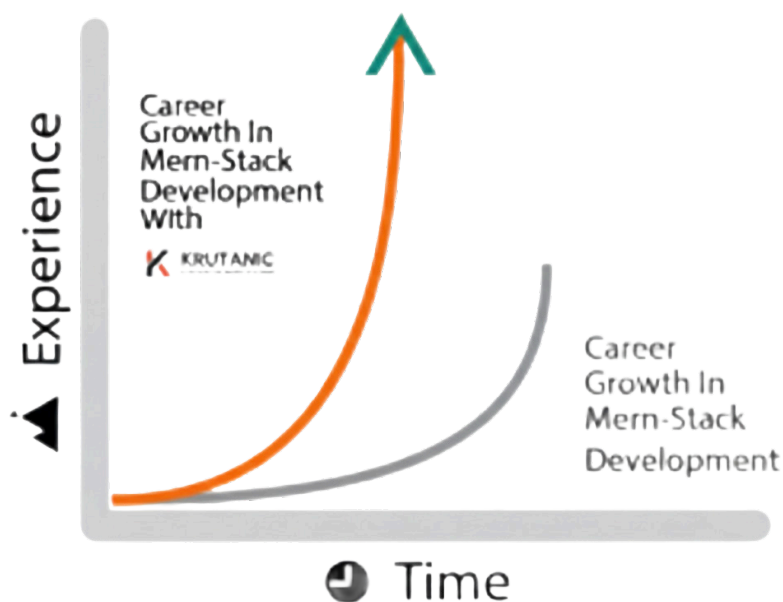
Build Portfolio

Learn and perfect your skills hands on by working on industry relevant projects to build a solid portfolio

Why is MERN-STACK DEVELOPMENT a Good career option?

MERN-STACK DEVELOPMENT is ever evolving & You need to evolve with it!

The global market size for MERN stack development is indirectly tied to the broader web development and software services industry, which is projected to grow significantly. The full-stack development segment (including MERN) is driven by increasing demand for scalable, high-performing web applications and single-page applications (SPAs). While specific data on MERN's market size is not readily isolated, the global web development services market is expected to reach \$56 billion by 2030, showcasing rapid growth. The MERN stack, being a JavaScript based solution, benefits from its open source nature, scalability, and flexibility, making it a top choice for startups and enterprises looking to build modern web applications. Its adoption in high-demand industries and companies such as Netflix and Uber further underscores its market prominence.



\$56 BILLION

Global MERN-STACK
DEVELOPMENT market by 2030

#1 MOST

IN - DEMAND SKILLS

55000+

Job opening's in India

How will the **MERN-STACK DEVELOPMENT** will help you achieve your Dream?

01

Online Induction

Meet your tribe, participate in fun yet intense challenges and kick-start your transformative journey.

02

Meet Master the concepts of **MERN STACK DEVELOPMENT**

Build a strong foundation in Mern-Stack Development and learn the strategies used by industry leaders to become a mern-stack developer in 6 months.

03

Build proof of work

Work on industry-relevant assignments, participate in challenging projects to develop a portfolio that showcases your diverse set of skills.

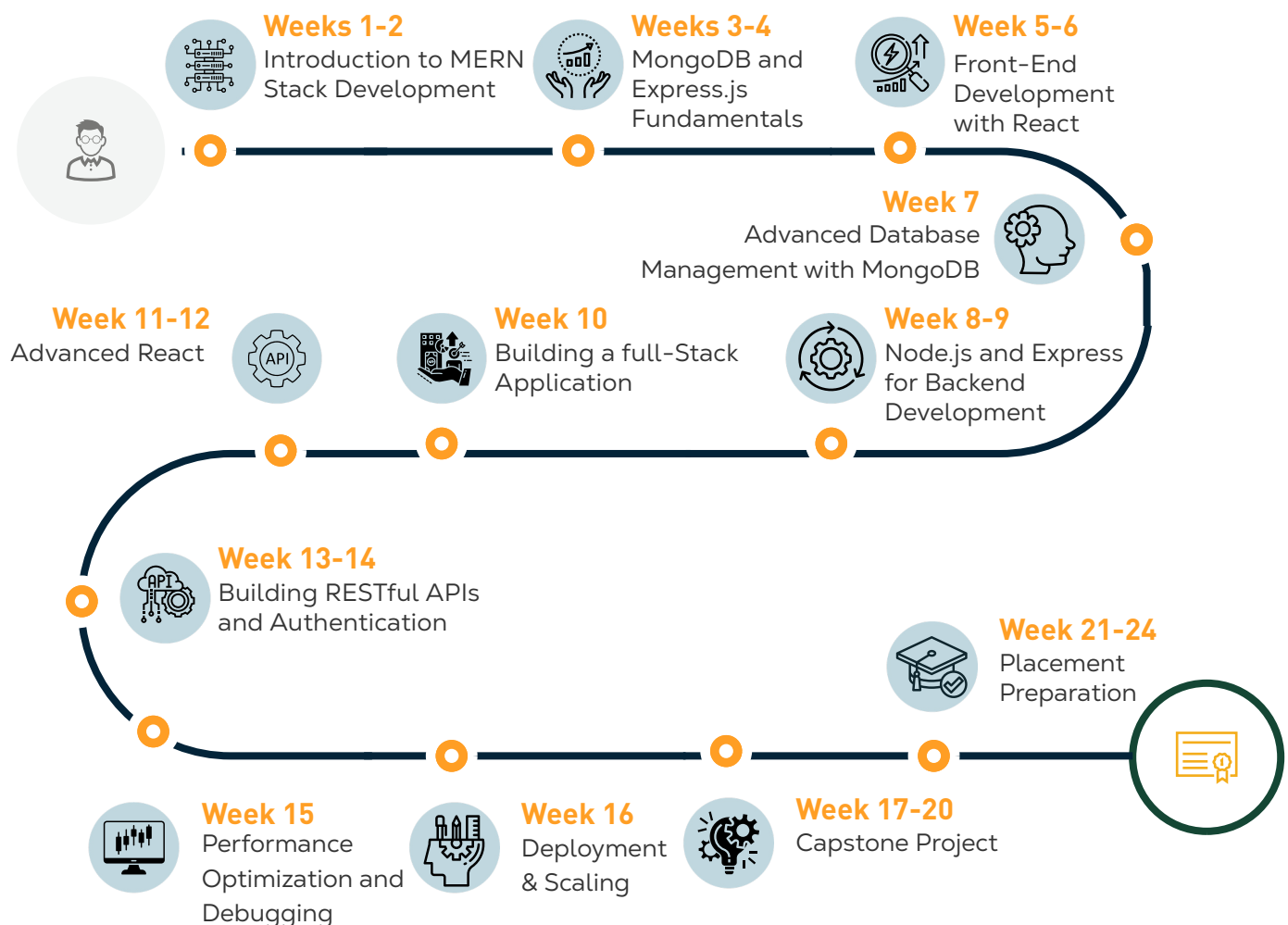
04

Prepare for interviews & land your dream job

Participate in live portfolio review sessions, mock interviews, and get guidance throughout the job application process.

Your Path to MERN STACK DEVELOPMENT Excellence

Step-by-Step Weekly Plan



Step-by-Step Weekly Plan

Week
1-2

1

2

Introduction to MERN Stack Development

Students will understand the MERN stack architecture and its individual components, enabling them to start building web applications.

- Overview of the MERN stack components
- (MongoDB, Express.js, React, Node.js)
- Setting up Node.js and MongoDB
- development environments
- JavaScript ES6+ for asynchronous programming
- Introduction to MongoDB databases

Learning Outcomes

Students will be introduced to the MERN stack (MongoDB, Express.js, React, Node.js) and its components. This module covers the basics of web development using JavaScript, understanding the architecture, and setting up the development environment.

Step-by-Step Weekly Plan

Week
3-4

3

4

MongoDB and Express.js Fundamentals

This module covers database management with MongoDB and building server-side applications using Express.js. It includes practical exercises for setting up databases, performing CRUD operations, and understanding the Express.js framework.

- MongoDB schema design
- CRUD operations in MongoDB
- RESTful APIs with Express.js
- Routing and middleware in Express

Learning Outcomes

Gain hands-on experience in building scalable backend applications using Express.js, focusing on developing efficient and well-structured RESTful APIs. Learn how to design, implement, and manage API endpoints while handling routing and request-response cycles effectively. Develop a strong understanding of middleware and its role in processing requests, managing authentication, error handling, and improving application performance.

Step-by-Step Weekly Plan

Week
5-6

5

6

Front-End Development with React

Students will focus on front-end development using React. This module covers the building of interactive user interfaces, managing state with React, and using JSX.

- React component lifecycle
- State management with React hooks
- Building UI components and React Router for navigation
- Using React libraries for animation and transitions

Learning Outcomes

Develop the ability to build interactive, responsive, and dynamic user interfaces using React, focusing on creating seamless user experiences. Gain a strong understanding of state management by effectively using React hooks such as `useState`, `useEffect`, and others to manage application data and lifecycle events. Learn component lifecycle methods and best practices to create reusable, modular, and maintainable components. Additionally, acquire skills in implementing efficient routing within React applications, enabling smooth navigation and scalable front-end architecture suitable for real-world projects.

Step-by-Step Weekly Plan

Week
7

7

Advanced Database Management with MongoDB

Students will master setting up and managing a Node.js server and understand asynchronous programming using advanced MongoDB techniques for optimizing queries and data management.

- Advanced MongoDB query techniques (aggregation, indexing, sharding)
- Data modeling and performance tuning
- Real-time data processing with MongoDB change streams
- Backup and restore strategies
- Database security practices

Learning Outcomes

They will understand how to secure databases and ensure performance tuning for complex applications, providing expertise in handling real-time data updates and backups.

Step-by-Step Weekly Plan

**Week
8-9**

8

9

Node.js and Express for Backend Development

This module focuses on back-end development using Node.js. Topics include server setup, handling HTTP requests, asynchronous operations, and API development with Express.js.

- Asynchronous programming in Node.js
- Creating APIs in Express.js
- Error handling and file management
- Using middleware effectively
- Implementing sessions and cookies

Learning Outcomes

Students will master setting up and managing a Node.js server and understand asynchronous programming.

They will be skilled in building robust RESTful APIs using Express.js and managing user sessions through authentication and cookies, ensuring secure data transmission.

Step-by-Step Weekly Plan

Week
10

10

Building a full-Stack Application

Students will integrate front-end React with backend Express to build a full-stack application. The focus is on data flow between the front-end and back-end and deploying the application.

- Connecting React with Express
- API integration and data flow management
- Deployment strategies (Heroku, AWS)
- Using axios for HTTP requests

Learning Outcomes

Students will be capable of building and integrating full-stack applications, handling API calls, and setting up deployments on cloud platforms. They will understand data flow between front-end and back-end components and implement strategies for scaling and monitoring applications in production.

Step-by-Step Weekly Plan

Week
11-12

11

12

Advanced React

This module covers advanced topics in React, including hooks, context API, custom hooks, and building complex applications.

- Use of React hooks and context API
- Custom hooks for data fetching
- Component composition and performance
- optimization
- Handling complex state management

Learning Outcomes

Students will gain expertise in using React hooks and context API, allowing for better state management and building more complex applications.

They will optimize React components for performance, understanding best practices for handling state across different components.

Step-by-Step Weekly Plan

Week
13-14

13

14

Building RESTful APIs and Authentication

This module focuses on building secure RESTful APIs and implementing authentication mechanisms using JWT.

Students will explore OAuth and secure data handling.

- Secure RESTful API building
- Implementing JWT for authentication
- OAuth integration
- API security best practices

Learning Outcomes

Students will be proficient in building secure APIs using RESTful methods and implementing JWT for authentication.

They will understand OAuth integration and secure data handling, allowing for the development of safe and effective web applications.

Step-by-Step Weekly Plan

Week
15

15

Performance Optimization and Debugging

Gain a comprehensive understanding of techniques used to optimize the performance of applications, ensuring faster load times, efficient resource utilization, and a smooth user experience. Learn how to identify performance bottlenecks in both front-end and back-end systems through profiling and monitoring tools. Develop practical skills in debugging by analyzing errors, tracing issues, and implementing effective solutions using industry-standard debugging tools and practices.

- Profiling tools (Chrome DevTools, React DevTools)
- Identifying and resolving performance bottlenecks
- Debugging Node.js and React applications
- Best practices for optimization

Learning Outcomes

Students will be able to use profiling tools to optimize the performance of their applications and effectively debug front-end and back-end issues. They will develop skills for resolving performance bottlenecks and ensuring efficient, smooth-running applications.

Step-by-Step Weekly Plan

**Week
16**

16

Deployment and Scaling

This module covers deployment strategies for MERN applications on cloud platforms like AWS, Heroku, and Firebase. Students will learn CI/CD pipelines and how to scale applications.

- CI/CD with GitHub Actions or Jenkins
- Containerization with Docker
- Deploying to cloud platforms (AWS, Heroku, Firebase)
- Scaling strategies and monitoring

Learning Outcomes

Students will gain experience in deploying MERN applications using CI/CD pipelines, containerization, and cloud platforms.

They will understand scaling strategies, ensuring their applications can handle larger user bases and performance demands.

Step-by-Step Weekly Plan

Week
17-20

17

20

Capstone Project

This module allows students to apply what they've learned in a real-world project. It also includes career preparation elements like resume building, mock interviews, and job search strategies.

- Capstone project development
- Portfolio building and resume preparation
- Mock interviews and job search strategies
- Networking with industry professionals

Learning Outcomes

Students will complete a comprehensive, end-to-end project that showcases their proficiency in MERN stack development, including MongoDB, Express.js, React, and Node.js. They will gain hands-on experience in designing, developing, and deploying full-stack applications that integrate both front-end and back-end technologies. Through this project, students will strengthen their problem-solving abilities, apply best practices in coding and architecture, and build scalable, real-world solutions.

Step-by-Step Weekly Plan

Week
21-24

21

24

Placement Preparation

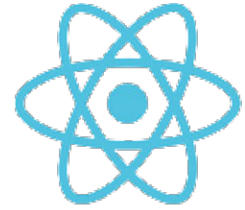
Focus on comprehensive career readiness by equipping students with essential skills required for successful job placements. Learn how to create impactful, professional resumes that effectively highlight technical skills, projects, and achievements. Gain hands-on experience through mock interviews, including technical and HR rounds, to build confidence and improve communication skills.

- Resume Building and LinkedIn Optimization
- Interview Tips and Mock Interviews
- Job Search Strategies and Networking
- How to Present Your Capstone Project
- Building a Professional Portfolio

Learning Outcomes

Prepare for a successful job search and interviews with a professional portfolio and job-ready skills. They will be prepared for full-stack developer roles, equipped with interview skills, resume-building techniques, and strategies for job placement.

Mern-Stack Development TOOLS



141000+ Job opening's worldwide



Full-Stack Developer



Frontend Developer



Backend Developer



MERN Stack Developer Web



Web Application Developer



React Developer



Node.js Developer



Technical Consultant



JavaScript Developer



Software Engineer



Mobile App Developer



Freelance Web Developer



DevOps Engineer



API Developer



UI/UX Developer

Will I get a certificate to enhance my resume?

Upon completing the **MERN STACK DEVELOPMENT Advanced Program**, you'll earn a certificate, highlighting your skills and expertise.



SHOWCASE
YOUR NEW ABILITIES
AND BUILD A STRONGER
PROFESSIONAL PORTFOLIO

All MERN-STACK DEVELOPMENT Concepts, Strategies, Assignments, Tools & Community Events At One Affordable Price

60,000/-

(+ 18% GST) 24 weeks | 60 Seats per cohort

- Access to an Exclusive Community of Top Marketers and Industry Professionals
- One-Year Access to Comprehensive Course Materials, Including Pre-reads and Training Resources.
- Exclusive Invitations to Physical Events, Networking Meetups, and Workshops.
- Personalized Guidance from Guest Mentors with Experience in Leading Startups.
- Real-World Insights and Strategies to Accelerate Your Marketing Career.

OUR ALUMMI'S

Top clients we work with:



Ready to Take Charge of Your Career?

Take your first steps towards a rewarding career in **MERN-STACK DEVELOPMENT** by filling out application form



**FINALLY PROGRAM KICK
OFF & ON BOARDING**



APPLY NOW



Got more question for us? Feel free to reach out to us at www.krutanic.com



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